

Arduino Timer Interrupts by

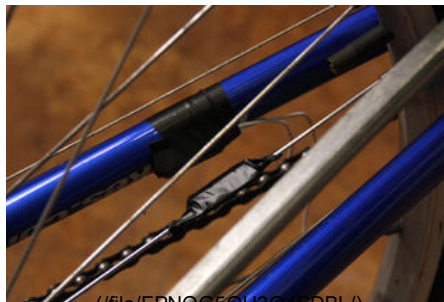
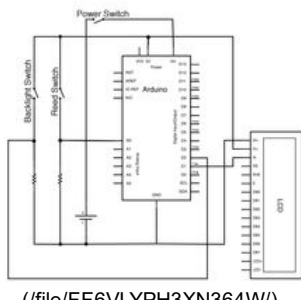
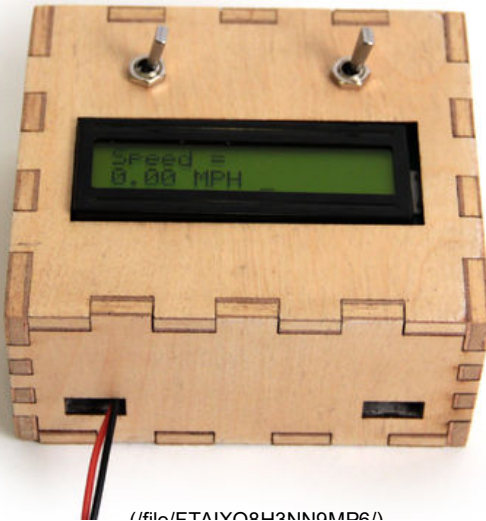
[amandaghassaei \(/member/amandaghassaei/\)](#) in [arduino \(/tag/type-id/category-technology/channel-arduino/\)](#)

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6 Steps

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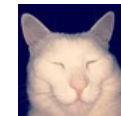
Step 3: Example 1: Bike Speedometer



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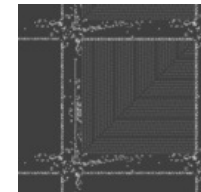
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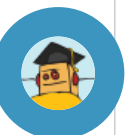
Bio: I'm a grad student at the Center for Bits and Atoms at MIT Media Lab. Before that I worked at Instructables, writing code for ... [More » \(/member/amandaghassaei/\)](#)

More by amandaghassaei:

[\(/id/OTCA-](#)[Metapixel-Conways-Game-of-Life/\)](#)[\(/id/9-Degrees-of-](#)[Freedom-IMU/\)](#)[\(/id/Twitter-Controlled-Pet-Feeder/\)](#)

In this example I made an arduino powered bike speedometer (<https://www.instructables.com/id/Arduino-Bike-Speedometer/>). It works by attaching a magnet to the wheel and measuring the amount of time it takes to pass by a magnetic switch mounted on the frame- the time for one complete rotation of the wheel.

I set timer 1 to interrupt every ms (frequency of 1kHz) to measure the magnetic switch. If the magnet is passing by the switch, the signal from the switch is high and the variable "time" gets set to zero. If the magnet is not near the switch "time" gets incremented by 1. This way "time" is actually just a measurement of the amount of time in milliseconds that has passed since the magnet last passed by the magnetic switch. This info is used later in the code to calculate rpm and mph of the bike.



Here's the bit of code that sets up timer1 for 1kHz interrupts

```
cli();//stop interrupts
//set timer1 interrupt at 1kHz
TCCR1A = 0;// set entire TCCR1A register to 0
TCCR1B = 0;// same for TCCR1B
TCNT1 = 0;//initialize counter value to 0
// set timer count for 1khz increments
OCR1A = 1999;// = (16*10^6) / (1000*8) - 1
//had to use 16 bit timer1 for this bc 1999>255, but could switch to timers 0 or 2
with larger prescaler
// turn on CTC mode
TCCR1B |= (1 << WGM12);
// Set CS11 bit for 8 prescaler
TCCR1B |= (1 << CS11);
// enable timer compare interrupt
TIMSK1 |= (1 << OCIE1A);
sei();//allow interrupts
```

Here's the complete code if you want to take a look:

```
//bike speedometer
//by Amanda Ghassaei 2012
//https://www.instructables.com/id/Arduino-Timer-Interrupts/
//https://www.instructables.com/id/Arduino-Timer-Interrupts/

/*
 * This program is free software; you can redistribute it and/or modify
 * it under the terms of the GNU General Public License as published by
 * the Free Software Foundation; either version 3 of the License, or
 * (at your option) any later version.
 */

//sample calculations
//tire radius ~ 13.5 inches
```

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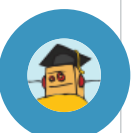
AravindP4 (/member/AravindP4)

8 days ago

Reply

thank u sir

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andreondra (/member/andreondra)





Great instructable! Thank you very much!

5 months ago

[Reply](#)



MalachiB1 (/member/MalachiB1)

a year ago

[Reply](#)

Thank you for breaking this down. I've read a lot of articles about timers but only yours makes it clear!



dineshc31 (/member/dineshc31) ▶ [MalachiB1](#) (/member/MalachiB1)

[Reply](#)

how exactly this is working for you all ?

6 months ago

For me while compiling in arduino 1.6.10 it is showing

error: 'TIMSK1' was not declared in this scope which is same for all above registers. do we need any #include header file here?



KennyL15 (/member/KennyL15)

9 months ago

[Reply](#)

Should also mention that the clock prescaler bit names CSX0, CSX1, CSX2 has the first number as 'X'. That number 'X' is linked to the timer number. So if 'X' is 0, then CS00, CS01, CS02 are prescaler bit names for timer 0; and the second value represents the bit position. So C00 means bit 0 (right-most-bit) of the 8-bit register TCCR0B, which is for timer 0.

The reason why I'm indicating this is because the tables (diagram) that go with "prescalers and compare match register" is confusing - since there's no mention about what CS10, CS11, CS12 etc are actually related to.



KennyL15 (/member/KennyL15)

9 months ago

[Reply](#)

"so a counter will take 10/16,000,000 seconds to reach a value of 9 (counters are 0 indexed)"

Do you actually mean it takes 10/16000000 seconds to begin (initial state) from index 0 and get back to a 0 index again? That is, initial index of 0. Then first increment is 1, second increment is 2, third increment is 3, all the way to 9th increment to index 9, and then one extra increment to get back to index 0 again. Altogether 10 increments to 'get back to 0'.



PeterMortensen (/member/PeterMortensen)

11 months ago

[Reply](#)

The comment for the line containing "ISR(TIMER2_COMPA_vect)" is confusing. "**timer1**" should probably be "**timer2**".



Sharmila07 (/member/Sharmila07)

a year ago

[Reply](#)

my signal is in the frequency range between 20Hz and 500Hz, so i am taking the sampling frequency as 5000Hz (sampling time will be .2ms), after sampling i need to store the sampled values in the SD card and so i need to use spi. i want to know how much time will spi take to store in SD card and will i be able to sample my signal at this rate? kindly help



ParthJ5 (/member/ParthJ5)

a year ago

[Reply](#)

Can anyone please explain the connections to arduino in example 1 ?
I want to know how is the signal generator and Oscilloscope is connected to the Arduino ?

Thanks



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Bruce McCarthy (/member/Bruce McCarthy)





Brilliant! I struggled with the datasheets for a few days before a year ago finding this instructable, and now it's all clear.

Reply

Thanks for sharing.



avionics2 (/member/avionics2)

a year ago

Reply

This is the best interrupts explanations I have come across.

Thanks !



jerrychee94 (/member/jerrychee94)

a year ago

Reply

Hi, currently im doing project where it prompt user in serial communication and doing interrupt ISR at the same time. So , when user enter a specific value , it will stop the interrupt for few seconds and then restart the interrupt again. However , when restart the interrupt , the timing is out.(expected interrupt time : 5 microsec per interrupt , actual interrupt time : higher than 5 microsec)

my code is here..

void setup()

{

cli();

TCCR0A = 0;

TCCR0B = 0;

TCNT0 = 0;

OCR0A = 80;// = (16*10^6) / (200000*1) - 1 (must be <256)

//200000hz=5microsec

TCCR0A |= (1 << WGM01);

TCCR0B |= (0 << CS02) | (0 << CS01) | (1 << CS00);

TIMSK0 |= (1 << OCIE0A);

sei();

}

void loop()

{

if(serial.available>0)

{

//if user press something...

TIMSK0 &= ~(1 << OCIE0A);

delay(10000);

TIMSK0 |= (1 << OCIE0A);

}

}

ISR(TIMER0_COMPA_vect)

{

//code

}

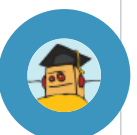


PiraC (/member/PiraC)

a year ago

Reply

Good Work. Thank you!





```
// turn on CTC mode
TCCR0A |= (1 << WGM01);
// Set CS01 and CS00 bits for 64 prescaler
TCCR0B |= (1 << CS01) | (1 << CS00);
```

a year ago [Reply](#)

What is the TCCR0A register? TCCR0B must be the 3-bit one as illustrated (where you set the prescaler) but I don't recall any mention of the other one.



MichealM1 (/member/MichealM1)

a year ago [Reply](#)

This made it much clearer than other reads I have been to. Based on this explanation, I was able to devise a routine to discipline my software clock based on the 16MHz crystal. One of my units (Uno) is ticking at a very wrong speed. Gaining about 30 seconds per day. That's a LOT! But the software can now take care of it and any other errant crystal it encounters. Thanks!!



MichealM1 (/member/MichealM1) ▶ MichealM1 (/member/MichealM1)

[Reply](#)

Sorry. Left out that I am getting NTP time hacks once per day but losing 30 seconds per day is way overboard. Maybe I will replace the crystal and see but as long as I can correct for it, it's not so bad.

a year ago



Baguda (/member/Baguda) made it!

a year ago [Reply](#)

In the code you set the timer1 register to CTC mode by "TCCR1B |= (1 << WGM12);". However, for all of the examples using timer0 and timer2, you set the CTC mode by "TCCR0B |= (1 << WGM01);" why do you set bit1 high instead of bit2 for timers 0 and 2 to put them into CTC mode? I found this table online that lists the mode settings(see attached), and it states that (100) is CTC mode. However, it is only for timer1... Is it different for timers 0 and 2?

Timer	Mode	WGM	CS	CFR	Mode	WGM	CS	CFR
0	0	0	0	0	Normal	0	0	0
0	0	0	1	0	Normal	0	1	0
0	0	0	1	1	Normal	0	1	1
0	0	1	0	0	Normal	1	0	0
0	0	1	0	1	Normal	1	0	1
0	0	1	1	0	Normal	1	1	0
0	0	1	1	1	Normal	1	1	1
0	1	0	0	0	Normal	0	0	0
0	1	0	0	1	Normal	0	0	1
0	1	0	0	1	Normal	0	0	1
0	1	0	1	0	Normal	0	1	0
0	1	0	1	1	Normal	0	1	1
0	1	1	0	0	Normal	1	0	0
0	1	1	0	1	Normal	1	0	1
0	1	1	1	0	Normal	1	1	0
0	1	1	1	1	Normal	1	1	1
1	0	0	0	0	Normal	0	0	0
1	0	0	0	1	Normal	0	0	1
1	0	0	1	0	Normal	0	1	0
1	0	0	1	1	Normal	0	1	1
1	0	1	0	0	Normal	1	0	0
1	0	1	0	1	Normal	1	0	1
1	0	1	1	0	Normal	1	1	0
1	0	1	1	1	Normal	1	1	1
2	0	0	0	0	Normal	0	0	0
2	0	0	0	1	Normal	0	0	1
2	0	0	1	0	Normal	0	1	0
2	0	0	1	1	Normal	0	1	1
2	0	1	0	0	Normal	1	0	0
2	0	1	0	1	Normal	1	0	1
2	0	1	1	0	Normal	1	1	0
2	0	1	1	1	Normal	1	1	1
2	1	0	0	0	Normal	0	0	0
2	1	0	0	1	Normal	0	0	1
2	1	0	1	0	Normal	0	1	0
2	1	0	1	1	Normal	0	1	1
2	1	1	0	0	Normal	1	0	0
2	1	1	0	1	Normal	1	0	1
2	1	1	1	0	Normal	1	1	0
2	1	1	1	1	Normal	1	1	1

(<https://cdn.instructables.com/FK7/FFSP/IJIUVOFI/FK7FFSPIJIUVOFI.LARGE.jpg>)



AnthonyP7 (/member/AnthonyP7)

2 years ago [Reply](#)

If the Arduino is executing some code within an interrupt does the counter continue to count up so that the next interrupt is correctly timed?



sandro2204 (/member/sandro2204) ▶ AnthonyP7 (/member/AnthonyP7)

[Reply](#)

Yes it does. But you should keep your interrupt service routine (ISR) as short as possible. Store a variable, set a flag and that's it. Do time consuming stuff in your main loop.

a year ago



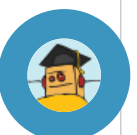
hal4512 (/member/hal4512)

a year ago [Reply](#)

according to datasheet of atmega328p p106 -108 when we choose values of COM0A0 and COM0B0 for timer0

we set the behavior of pins OC0A OCOB After having config them as output and hardware will proceed as its describe in the datasheet.

all modes Normal mode ,CTC mode non PWM ..ect ..are described in page 100



when you'll read: For generating a waveform output in CTC mode the OC0A output can be set to toggle each logical level on each compare match (you don't have to toggle it by soft).



hal4512 (/member/hal4512)

a year ago

[Reply](#)

you don't need to write a code for toggling because atmega328p has 3 timers and every timer has 2 output pins ex: timer2 has OC2A pin 11 and OC2B pin 3 and in isr routine u can do other thing.



KevinK91 (/member/KevinK91) ▶ hal4512 (/member/hal4512)

a year ago

[Reply](#)

could you please elaborate more on why the toggle code isn't necessary in this sketch?



hackerh (/member/hackerh)

a year ago

[Reply](#)

so if you wanted an interrupt every second (frequency of 1Hz):
compare match register = $[16,000,000 / (\text{prescaler} * 1)] - 1$
with a prescaler of 1024 you get:



metrog (/member/metrog)

a year ago

[Reply](#)

Hi,

Your project and explanations helped me a lot starting with my project. Thanks!

I'm using Arduino uno timer 0 and 2 to output 2 distinct (in sync) MIDI Clock signal and it works like charm. I can change BPM with a knob going from 30 to 250.

I'm struggling on my next step which is introducing an offset (-200ms / +200ms) between the 2 signals using an analogRead on a knob.

Would you know how to induce such an offset for a Serial output?

Timr0 and Timer2 doesn't have much room to change start point and end point and the induced delay would only be a few milliseconds maybe.

Well if this message finds you and you wish to reply, I'm all ears opened.

Thanks again,

Stef



tejonbiker (/member/tejonbiker)

a year ago

[Reply](#)

Wonderful, I worked with PICs but I see the AVR work slightly different



youraagh (/member/youraagh)

2 years ago

[Reply](#)

Hi, great tutorial, thanks heaps.

What libraries do I need to include, I'm getting compiler errors saying it doesn't know the names of the timing registers.



PravinahS (/member/PravinahS)

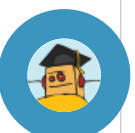
2 years ago

[Reply](#)

Hi Amanda,

I found these article very informative. I'm designing a card reader and i have included these timer interrupts in my programs. I have several confusions to use these timer interrupts on Arduino Leonardo. I have attached pins on pin 3 and pin 2 for D0 and D1 from my Mifare card reader. My card is a Mifare (32bits)

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type, so when i batch these card,i'm getting the extra bit of 33,so i decided to use timer interrupts just to execute exact 32 bits,but still i'm getting these extra bit.Please help me..Im lost!!



SitthinutK (/member/SitthinutK)

2 years ago

[Reply](#)

Thanks!!



RosmiR (/member/RosmiR)

2 years ago

[Reply](#)

In an Arduino code for pulse sensor 256 prescaler is activated by TCCR2b=0x06 . Why is it so ? Why is it not 0x04 as per the table for Timer0. Is it possible to make an interrupt of 100 Hz using Timer 2. (ocr2a <256, 1024 prescaler is not present). Could you help me with this. I am just a beginner.



RosmiR (/member/RosmiR)

2 years ago

[Reply](#)

How could I make an interrupt of 100Hz with timer 2. Could you share the table for timer2 similar to the one for Timer0 .



ProfRobNewton (/member/ProfRobNewton)

2 years ago

[Reply](#)

Just wondering if it would not be easier and more accurate to generate an interrupt when a switch change is registered and use millis() to determine the time since the last switch change? Sampling every 1ms using the method above may miss a switch change??



andycomiC (/member/andycomiC) ▶ [ProfRobNewton \(/member/ProfRobNewton\)](#)

2 years ago

[Reply](#)

I am no expert but to my knowledge the problem with switch state interrupts is that they bounce and your can't denounce a switch using software on an interrupt routine as its already been triggered (as in the interrupt routine is already done before you have a chance to de-bounce.). The other way is to use a hardware based button debounce switch which just uses a capacitor. This will solve the problem. However unless your project is huge, polling a switch every x cycles doesn't really make any difference to any project.



Fardenco (/member/Fardenco) ▶ [andycomiC \(/member/andycomiC\)](#)

[Reply](#)

2 years ago

I had the problem with a motor encoder, just put a 10k Ohms resistor to Vcc and a 1µF capacitor to GND around your interrupt pin, this will be ok !!

Pin interrupt would be more efficient because you'll never miss a change, and above that, you'll not waste time with useless interruption =)



DavidS13 (/member/DavidS13) ▶ [ProfRobNewton \(/member/ProfRobNewton\)](#)

2 years ago

[Reply](#)

While I don't think sampling every 1ms may miss a switch change (think about it, 1000 times a second???) I too would like to know if the more efficient way would be to make an interrupt for the switch change. I have a project that would use something similar and I'm trying to determine the most efficient course of action.



manoch_dc (/member/manoch_dc)

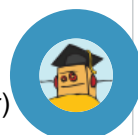
2 years ago

[Reply](#)

thank for Example interrupts that working.

!!But void loop() has not work

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```

my increase program

void setup(){
  //set pins as outputs
  pinMode(7, OUTPUT); //my led
  pinMode(8, OUTPUT);
  pinMode(9, OUTPUT);
  pinMode(13, OUTPUT);
  cli();//stop interrupts
  same example
  sei();//allow interrupts
}

void loop(){
  digitalWrite(7, HIGH); // turn the LED on (HIGH is the voltage level)
  delay(100); // wait for a second
  digitalWrite(7, LOW); // turn the LED off by making the voltage LOW
  delay(100); // wait for a second
  //do other things here
}

```



coltonschlosser (/member/coltonschlosser)

2 years ago

Reply

I made a python script that walks you through setting up the code for timer interrupts.
<https://gist.github.com/cltnschlosser/e747efae07a76e117e02>



killercatfish (/member/killercatfish)

2 years ago

Reply

Could this be used for a 2 player reflex timer game? Trying to push your players button on multiples of 5 seconds and getting a point when you do.



zinedine (/member/zinedine)

2 years ago

Reply

Good job, I want to import 200khz frequency, it's about 5us, but I only have 13.6us with every OCRnA that I used to change from 0 to 80. I'm using 2560. How can I do that?

Thanks



faizal91 (/member/faizal91)

2 years ago

Reply

Thank you... Very nice tutorial!



EnekoL (/member/EnekoL)

2 years ago

Reply

Hei, very usefull application, but i've one cuestion, how can i call de interrump for timer 1 inside the loop??Thanks!



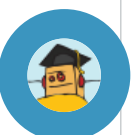
ohoilett (/member/ohoilett) ▶ **EnekoL** (/member/EnekoL)

2 years ago

Reply

I could be wrong. You may want to check

<http://arduino.cc/en/Reference/AttachInterrupt>





ohoilett (/member/ohoilett) ▶ EnekoL (/member/EnekoL) 2 years ago [Reply](#)

You don't call interrupts like normal functions. An interrupt service routine (ISR) is "called" or better yet "triggered" at the specified frequency that you determined when you set up the interrupt.



thegoodhen (/member/thegoodhen) 3 years ago [Reply](#)

Hello... The code is very useful, however, be aware that the sketch you posted doesn't seem to ever get into the main loop!



DavidS13 (/member/DavidS13) ▶ thegoodhen (/member/thegoodhen) 2 years ago [Reply](#)

I'm kind of a noob about all this, but unless I'm mistaken the point of interrupts is that you don't NEED a main loop. The main loop just runs forever and everything is done when interrupts happen. It is an extremely efficient way to code....



Unidentified Identity (/member/Unidentified Identity) 3 years ago [Reply](#)

can we generate square wave with varying pulse width with this interrupt ?



amandaghassaei (/member/amandaghassaei) (author) ▶ Unidentified Identity (/member/Unidentified Identity) 3 years ago [Reply](#)

yes it's possible, but probably easier to just use the analogWrite command



srinivas_arduino (/member/srinivas_arduino) 3 years ago [Reply](#)

sir please send program for generating 15000 sample values in 10 sec using timer interrupt values adc values from accelerometer sensor



srinivas_arduino (/member/srinivas_arduino) 3 years ago [Reply](#)

sir please send program for generating 15000 sample values using timer interrupt values adc values from accelerometer sensor



AbdulW87 (/member/AbdulW87) 3 years ago [Reply](#)

i am confused about something in the code, the line

```
// turn on CTC mode
```

```
TCCR0A |= (1 << WGM01);
```

shouldnt it be TCCR0A |= (1 << WGM02); or am i understanding this wrong, i am a beginner at arduino so its still very confusing to me.



O-Zo (/member/O-Zo) 3 years ago [Reply](#)

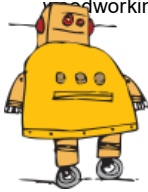
Thank you for this tutorial, I'm struggling with a project and this helped me stumble at least to the right direction.

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